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Can microaeration and hydraulic retention time steer anaerobic microbiomes toward efficient linear alkylbenzene sulfonate degradation?

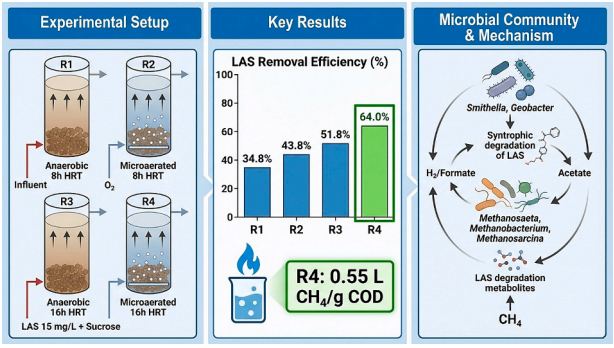
Maurício Guimarães de Oliveira, Vicente Elício Porfiro Sales Gonçalves da Silva, Steven Renato Ferreira Vasconcelos, Paulo Igor Milen Firmino, André Bezerra dos Santos *

Department of Hydraulic and Environmental Engineering, Federal University of Ceará, Fortaleza, Ceará, Brazil

HIGHLIGHTS

- Microaeration combined with 16-h HRT increases LAS removal to 64%.
- The 16-h HRT prevents washout of specialized microaerophilic consortia.
- Specific methane yield reached 0.28 L_{CH4}/g_{CODapp}, doubling the anaerobic control.
- High *Thauera* and *Desulfovibrio* abundances are associated with enhanced LAS removal.
- Synergy between redox zones overcomes the kinetic barriers of surfactant toxicity.

GRAPHICAL ABSTRACT



ARTICLE INFO

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ABSTRACT

The sustainable treatment of surfactant-rich wastewater in anaerobic systems is often hindered by the biochemical recalcitrance of linear alkylbenzene sulfonate (LAS) and its inhibitory effects on methanogenesis. This study investigated controlled microaeration under two hydraulic retention time (HRT) and organic loading rate (OLR) regimes as a strategy to intensify LAS degradation and energy recovery in upflow anaerobic sludge blanket (UASB) reactors. Four configurations were evaluated: R1 (anaerobic, 8-h HRT), R2 (microaerated, 8-h HRT), *R*3 (anaerobic, 16-h HRT), and R4 (microaerated, 16-h HRT). Results demonstrated that under strictly anaerobic conditions, operating at a 16-h HRT and the corresponding lower OLR resulted in higher chemical oxygen demand (COD) removal than at 8-h HRT (87.4% vs. 65.7%). In contrast, microaeration played an important role in enhancing surfactant biotransformation. R4 achieved the highest LAS removal (64.0 ± 6.2%), significantly outperforming the strictly anaerobic *R*3. The 16-h HRT/lower-OLR configuration provided operational conditions that may have favored the progression of successive LAS biotransformation steps, including oxygenase-mediated ω-oxidation and the transformation of aromatic compounds under microaerated conditions. Notably, R4 exhibited a remarkable specific methane yield of 0.28 ± 0.01 L_{CH4}/g_{CODapp}, doubling the efficiency of the other reactors. Microbial analysis indicated that microaeration was associated with a more diverse consortium in which potentially oxidative bacteria (*Smithella* and *Thauera*) coexisted with acetoclastic methanogens (*Methanosaeta*). These findings indicate that microaeration, when combined with the longer HRT tested,

* Corresponding author at: Department of Hydraulic and Environmental Engineering, Campus do Pici, Bloco 713, Pici. CEP: 60455-900, Fortaleza, Ceará, Brazil.
E-mail address: andre23@ufc.br (A.B. dos Santos).

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enhanced LAS degradation and process performance compared with the 8-h condition, highlighting the potential of this strategy for process intensification in treatment plants.

1. Introduction

Linear alkylbenzene sulfonate (LAS) is a widely used anionic surfactant that is frequently detected in domestic wastewater at concentrations of 0.8–21 $mg{\cdot}L^{-1}$ (Costa et al., 2024; Motteran et al., 2018). Its incomplete removal in conventional treatment plants results in environmental accumulation and adverse effects, including foam formation and impaired oxygen transfer, while its surfactant properties may also disrupt microbial membranes and inhibit biological activity at elevated concentrations (Luo et al., 2023; Teran et al., 2022). In upflow anaerobic sludge blanket (UASB) reactors, LAS removal typically ranges from 30% to 91%, a process that is highly dependent on the availability of fermentable co-substrates, such as ethanol or sucrose, to sustain syntrophic pathways and mitigate surfactant-induced metabolic stress (Costa et al., 2024; Terreros-Mecalco et al., 2024).

Despite these reported efficiencies, anaerobic LAS biodegradation remains challenging because its amphiphilic structure combines a hydrophobic alkyl chain with a sulfonated aromatic moiety. Interactions of the hydrophobic moiety with sludge and cell membranes can restrict bioavailability and inhibit microbial activity, whereas degradation of the aromatic structure requires initial activation. Under strictly anaerobic conditions, this activation may occur via fumarate addition, followed by β-oxidation, desulfonation, and aromatic ring transformation (Cui et al., 2022; Lara-Martín et al., 2010). These kinetically constrained reactions, together with surfactant-induced microbial inhibition, constitute major bottlenecks for LAS biodegradation, particularly at short contact times (Terreros-Mecalco et al., 2024; Wu et al., 2020).

Hydraulic retention time (HRT) is therefore an important operational parameter, as it determines the contact time available for these complex biotransformation steps. Granatto et al. (2019) achieved 60 ± 30% LAS removal in an Expanded Granular Sludge Bed (EGSB) reactor at an HRT of 36 h. In contrast, Carosia et al. (2014) reported 48 ± 10% in an Anaerobic Fluidized Bed Reactor (AFBR) at 15 h. In UASB reactors, Okada et al. (2013) reported only 18% LAS removal at an HRT of 6 h. However, extending the HRT requires larger reactor volumes and may compromise the economic advantages of UASB technology, particularly for domestic wastewater treatment, where HRTs of 6–8 h are commonly used (Chernicharo et al., 2015). Thus, strategies that alleviate kinetic limitations without relying exclusively on prolonged residence times are particularly relevant for LAS treatment.

Microaeration represents a potential strategy because controlled oxygen supplementation can promote initial oxidative transformations of recalcitrant compounds while preserving predominantly anaerobic conditions and the associated potential for energy recovery. Reported oxygen doses range from 0.005 to 5 L $O_2{\cdot}L^{1}_{feed}{\cdot}d^{-1}$ depending on the treatment objective and substrate composition (de Oliveira et al., 2021; Nguyen and Khanal, 2018).

Previous studies have demonstrated that controlled oxygen supplementation can substantially enhance the biotransformation of recalcitrant compounds in predominantly anaerobic systems. For BTEX, Siqueira et al. (2018) reported removal efficiencies ranging from 55 to 82% under strictly anaerobic conditions, whereas microaeration at 0.5–2.0 $mL_{air}{\cdot}min^{-1}$ increased removal to above 83% for all compounds under the best-performing conditions. At an airflow rate of 1.0 $mL_{air}{\cdot}min^{-1}$, benzene removal increased from 54.9 to 83.9% and toluene removal from 67.3 to 88.6%, while ethylbenzene and xylenes also achieved removals above 85%. Notably, increasing the airflow rate to 2.0 $mL_{air}{\cdot}min^{-1}$ did not further improve BTEX removal, highlighting the importance of controlled rather than excessive oxygen supplementation. Similar benefits of microaeration have been reported for other recalcitrant organic compounds (do Nascimento et al., 2021; Carvalho et al., 2020), supporting its potential as a process-intensification strategy that facilitates otherwise kinetically constrained biotransformation steps while preserving the advantages of predominantly anaerobic treatment.

More recently, Ortiz-Ardila et al. (2024) demonstrated that microaeration enhanced the conversion of volatile D4 and D5 siloxanes to methane and simpler products, while reshaping the anaerobic microbial community. Despite the structural differences between siloxanes and LAS, these findings reinforce the broader concept that controlled oxygen supplementation can facilitate initial transformation steps of recalcitrant compounds while preserving downstream anaerobic processes, including methanogenesis.

From this structural perspective, microaeration offers a feasible means to alleviate one of the main bottlenecks in anaerobic LAS degradation by supplying limited amounts of molecular oxygen to initiate molecular activation while preserving predominantly anaerobic conditions. Oxygen can support mono- and dioxygenase-mediated reactions involved in terminal hydroxylation of the alkyl chain (ω-oxidation) and subsequent transformations of the aromatic moiety, generating more hydrophilic carboxylated intermediates that are more accessible to downstream biodegradation (Cui et al., 2022; Liu et al., 2024). Thus, rather than replacing anaerobic metabolism, microaeration may act as an oxidative priming step that facilitates the transformation of structurally recalcitrant LAS into intermediates more amenable to subsequent anaerobic metabolic reactions (Lara-Martín et al., 2010; Xu et al., 2023).

Beyond enzymatic activation, microaeration combined with fermentable co-substrates, such as sucrose, may promote functional selection within the microbial community. This process favors the enrichment of a specialized microaerophilic-syntrophic consortium, notably dominated by the phylum *Pseudomonadota* and genera such as *Pleomorphomonas* and *Desulfovibrio* (Buarque et al., 2019; do Nascimento et al., 2021). This specialized structure facilitates a coupled metabolic sequence: an initial oxidative attack on the LAS molecule, followed by anaerobic desulfonation and methanogenesis by oxygen-tolerant archaea such as *Methanobacterium*. Consequently, microaeration bridges the efficiency of aerobic pathways with the energy recovery of anaerobic digestion without compromising process stability (Costa et al., 2024).

Despite the documented benefits of microaeration in anaerobic systems, its effect on LAS treatment under different hydraulic and organic loading conditions remains poorly understood. Therefore, this study evaluated microaeration under two operationally distinct HRT/OLR regimes. Therefore, this study compared two operationally distinct HRTs, 8 and 16 h, under anaerobic and microaerated conditions. The 8-h HRT is a short residence time commonly used in high-rate UASB systems. In contrast, the 16-h HRT served as a longer-residence-time benchmark to assess whether extending substrate–biomass contact could alleviate kinetic limitations.

2. Materials and methods

2.1. Experimental set-up

Four lab-scale UASB reactors (working volume of 3.5 L) (Fig. 1), inoculated with anaerobic sludge [Total solids (TS) = 55.1 ± 3.5 $g{\cdot}L^{-1}$, total volatile solids (TVS) = 30.6 ± 1.4 = $g{\cdot}L^{-1}$ and total fixed solids (TFS) = 24.7 ± 1.1 $g{\cdot}L^{-1}$], were operated, in parallel, at a room temperature of approximately 28 °C.

The reactors were operated under the same conditions except for microaeration and HRT. R1 and *R*3 were conventional UASB reactors, while R2 and R4 included microaeration and were referred to as UMSB

(Upflow Microaerobic Sludge Blanket). Regarding hydraulic retention time, R1 and R2 had an HRT of 8 h, while *R*3 and R4 had an HRT of 16 h. The 16-h HRT was selected as a strategic benchmark to confirm that any performance limitations at 8 h were kinetic rather than due to inoculum or system instability. This approach is based on prior research by the group, which showed that short residence times led to only 32 ± 16% LAS removal in the microaerobic system (de Oliveira et al., 2026).

Microaeration was performed with synthetic air (80% N_2:20% O_2, White Martins, Brazil) from the bottom of the reactor, using a mass flow controller (GFC17, Cole-Parmer, USA). The microaeration air dose used in R2 and R4 was approximately 1.5 $mL_{air}{\cdot}min^{-1}$, corresponding to 0.041 e 0.082 L $O_2{\cdot}L_{feed}^{-1}{\cdot}d^{-1}$, respectively. The gas was introduced at the bottom of each reactor through a bubble diffuser, promoting its dispersion into the liquid phase along the upflow path. The microaeration dose was selected based on the best results reported by do Nascimento et al. (2021) for the removal of emerging sulfur-containing compounds.

The reactors were fed with synthetic wastewater containing the LAS (Sodium dodecylbenzenesulfonate, 100%, Sigma-Aldrich, USA) (15 $mg{\cdot}L^{-1}$), sucrose (1 g $COD{\cdot}L^{-1}$), basal medium (macro and micronutrients) prepared according to Firmino et al. (2010), and sodium bicarbonate (1 $g{\cdot}L^{-1}$) to maintain the pH close to 7.0. The organic loading rates (OLRs) were 3.0 g $COD{\cdot}L^{-1}{\cdot}d^{-1}$ for R1 and R2 (8-h HRT) and 1.5 g $COD{\cdot}L^{-1}{\cdot}d^{-1}$ for *R*3 and R4 (16-h HRT). All reagents were used as purchased without further purification. The synthetic influent was kept refrigerated at 4 °C to prevent natural biodegradation of organic matter.

2.2. Chemical and chromatographic analysis

Chemical oxygen demand (COD), pH, TSS, and TS content were determined according to APHA (2012).

The quantification of CH_4, CO_2, H_2, and H_2S in the biogas was determined by gas chromatography with ionization detection by dielectric barrier discharge (GC BID-2010 Plus, Shimadzu Corporation, Japan), equipped with a GS-GASPRO column (60 m × 0.32 mm) (Agilent Technologies Inc., USA). Helium gas (White Martins LTDA, Brazil) was used as the carrier gas at a flow rate of 2 $mL{\cdot}min^{-1}$, with a run time of 9 min. Oven, injector, and detector temperatures were 50, 100, and 250 °C, respectively. O_2 and N_2 were quantified by gas chromatography with thermal conductivity detection (GC-TCD) (GC-17A, Shimadzu Corporation, Japan). The biogas sample (1.0 mL) was injected in splitless mode, and chromatographic separation was performed on a Mol Sieve 5A PLOT column (30 m, 0.32 mm ID) (Restek Corporation, USA). Oven, injector, and detector temperatures were 35, 40, and 230 °C, respectively. Helium (White Martins, Brazil) was used as the carrier gas at a flow rate of 7 $mL{\cdot}min^{-1}$, and the run time was 5 min.

LAS concentrations were determined by high-performance liquid chromatography (HPLC) using a Shimadzu system (CBM-20A, SPD-20A, CTO-20A, SIL-20A, and LC-20AT) equipped with a reversed-phase C18 column from Shimadzu, as described by Silva et al. (2017). The mobile phase was water with 5 $mmol{\cdot}L^{-1}$ ammonium acetate (solvent A) and acetonitrile (solvent B); the mobile phase flow rate was 1 $mL{\cdot}min^{-1}$, the sample volume was 50 μL, the temperature of the thermostated compartment of the column was 40 °C, and UV–Vis detection at 230 nm. A gradient was applied to the mobile phase to separate the LAS constituents; it started at 45% solvent B, which began to rise after 5 min and reached 60% in 20 min, then returned to 45% after 1 min and remained at 45% for 6 min. Volumetric LAS removal rates were calculated from the difference between influent and effluent LAS concentrations, considering the influent flow rate and the working volume of each reactor, and were expressed as mg $LAS{\cdot}L_{reactor}^{-1}{\cdot}d^{-1}$.

2.3. Off-gas oxygen depletion analysis

To provide an independent assessment of oxygen depletion under microaeration, the O_2 and N_2 concentrations in the reactor off-gas were monitored throughout the operational period and used to calculate the off-gas O_2/N_2 ratio. Nitrogen was used as an internal tracer because it is expected to behave conservatively under the evaluated operational conditions, whereas oxygen may be transferred to the liquid phase and depleted through oxidative processes.

The apparent gas-phase oxygen depletion was calculated according

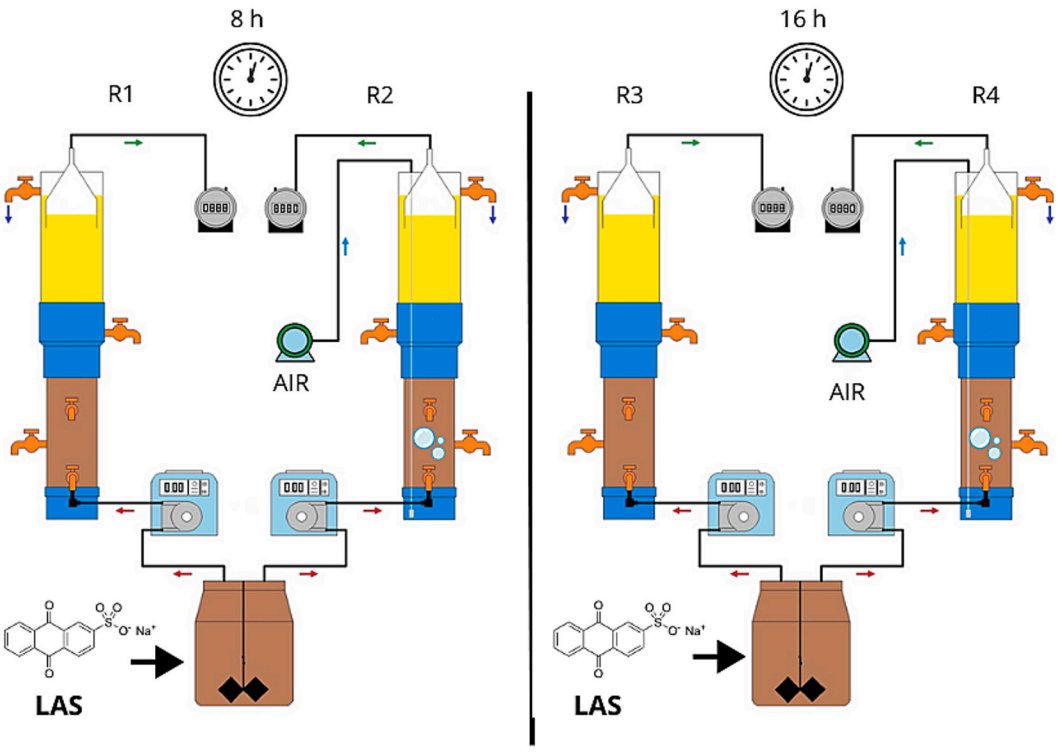


Fig. 1. Schematic of the continuous experimental set-up.

to Equation (1):

$$Apparent O_2 depletion(\%) = \left[1 - \frac{(O_2/N_2)_{off-gas}}{0.25}\right] \times 100$$

where $(O_2/N_2)_{off-gas}$ is the measured ratio in the reactor off-gas and 0.25 corresponds to the O_2/N_2 ratio of the supplied synthetic air. Normalization to N_2 minimizes the influence of dilution by biologically produced CH_4 and CO_2, which simultaneously dilute O_2 and N_2, whereas preferential O_2 depletion decreases their ratio. Although Oxidation-reduction potential (ORP) and dissolved oxygen (DO) were not monitored, this analysis provided an independent gas-phase assessment of the depletion of the supplied O_2 during reactor operation.

2.4. Statistical analysis

The Mann-Whitney and Kruskal-Wallis nonparametric tests, which do not require specific distributional assumptions, were used to compare reactor performance across the different experimental periods at the 5% significance level.

2.5. Microbiological analysis

Composite sludge samples were collected from all reactors at the beginning and end of each operational period. Total DNA was extracted from 0.25 g of fresh biomass using the PowerSoil® DNA Isolation Kit (MoBio Laboratories Inc., USA). The V4 region of the 16S rRNA gene was amplified in triplicate using specific primers 515F/806R. Post-amplification, products were purified with Agencourt AMPure XP beads (Beckman Coulter) and quantified using the Qubit 2.0 fluorimeter (Invitrogen). Sequencing was performed on an Illumina MiSeq platform using 300-cycle chemistry.

Bioinformatics processing was conducted using the DADA2 package v1.28.0 to filter and assemble reads into amplicon sequencing variants (ASVs) (Callahan et al., 2016). Taxonomic classification was assigned using the MiDAS database v5.1, which is specialized for anaerobic digestion consortia.

Alpha diversity indices were calculated using the Phyloseq package v1.44.0. Metabolic functional profiles were predicted from the 16S rRNA gene sequencing data using PICRUSt2 (Phylogenetic Investigation of Communities by Reconstruction of Unobserved States 2), and predicted gene families were annotated based on KEGG Orthologs (KOs). The resulting predictions were used to compare the potential representation of metabolic functions among the microbial communities of the different reactors.

Raw sequence data are available in the NCBI BioProject database under ID PRJNA836880.

3. Results and discussion

3.1. Performance of the UASB reactors

3.1.1. Operational parameters and volumetric biogas production

The monitoring of pH and Acidity/TA ratios showed no statistically significant differences between the reactors ($p > 0.05$), indicating that the system maintained a stable buffering capacity regardless of the HRT or the redox environment (Fig. S1). All reactors operated within an alkaline pH range (7.6–8.0), which is optimal for methanogenic activity.

The Acidity/TA ratios remained consistently above the 0.3 threshold (0.5–0.6) across all treatments. This absence of statistical variance suggests that the accumulation of volatile fatty acids (VFAs) was a common characteristic of the influent (sucrose + LAS) rather than a result of operational changes. The stability of these parameters demonstrates that neither the high organic load at 8-h HRT nor the oxygen supply in the microaerated systems (R2 and R4) compromised the global metabolic balance, thereby enabling subsequent evaluation of LAS removal under a stabilized anaerobic-microaerophilic environment.

The biogas dynamics reflected the combined influence of the operational conditions and redox environment (Fig. 2). Among the strictly anaerobic reactors, R1, operated at an 8-h HRT and the corresponding higher OLR, showed a higher volumetric methane production (1.2 ± 0.1 $L_{CH4}{\cdot}d^{-1}$) than *R*3 (1.0 ± 0.1 $L_{CH4}{\cdot}d^{-1}$), operated at a 16-h HRT and lower OLR ($p = 0.043$). Because the influent COD concentration was maintained constant, decreasing the HRT from 16 to 8 h by increasing the influent flow rate concomitantly increased the OLR from approximately 1.5 to 3.0 g $COD{\cdot}L^{-1}{\cdot}d^{-1}$. Therefore, the difference in volumetric methane production between R1 and *R*3 cannot be attributed solely to either HRT or OLR; rather, it reflects the combined effect of these coupled operational parameters.

In the microaerated reactors, R2 and R4 exhibited total gas flow rates of 9.6 and 4.7 $L{\cdot}d^{-1}$, respectively. These values include the externally supplied synthetic air and therefore cannot be interpreted exclusively as biologically produced biogas. Nevertheless, absolute methane production can be directly compared between the reactors and was approximately twofold higher in R2 than in R4 (2.6 vs. 1.4 $L_{CH4}{\cdot}d^{-1}$, $p < 0.001$). This difference is consistent with their hydraulic operation: because R2 was operated at half the HRT of R4 (8 vs. 16 h) at the same influent COD concentration, its daily influent flow and, consequently, COD load were approximately twice those of R4. Thus, the greater absolute methane production in R2 primarily reflects the greater amount of organic substrate supplied per unit time rather than evidence of a higher methane conversion efficiency.

Methane content also differed between the microaerated reactors. R4 (16 h) achieved $30 \pm 2\%$ CH_4, nearly reaching the 45% energy recovery threshold (de Oliveira et al., 2026), whereas R2 (8 h) exhibited a lower CH_4 content of $26 \pm 2\%$. Since both reactors received the same airflow rate, the higher CH_4 fraction in R4 indicates a more favorable balance between methane generation and the total gas flow under this operational condition. This gas quality is comparable to that observed by Buarque et al. (2019) under the same operating conditions.

More importantly, comparison within each HRT showed greater absolute methane production under microaeration. At 8 h, methane production increased from 1.2 ± 0.1 $L{\cdot}d^{-1}$ in R1 to approximately 2.6 $L_{CH4}{\cdot}d^{-1}$ in R2 ($p < 0.001$), whereas at 16 h it increased from 1.0 ± 0.1 $L_{CH4}{\cdot}d^{-1}$ in *R*3 to approximately 1.4 $L_{CH4}{\cdot}d^{-1}$ in R4 ($p = 0{,}022$). Because each anaerobic–microaerated pair was operated at the same HRT and corresponding OLR, these comparisons provide a more direct indication of the association between microaeration and methane production than comparisons across different HRTs. These results surpass the yields reported by Okada et al. (2013), who found that anaerobic reactors treating high LAS concentrations (up to 40 $mg{\cdot}L^{-1}$) exhibited significant methanogenic inhibition and lower volumetric production. The higher methane production observed under microaeration is consistent with the possibility that limited oxygen supplementation alleviated some of the constraints imposed by LAS on methanogenic activity, as also suggested for micro-oxygenated systems treating sulfonated aromatic compounds (Zhou et al., 2022).

3.1.2. Efficiency parameters

COD removal efficiency across all experimental systems ranged from $65.7 \pm 12.3\%$ to $87.4 \pm 3.4\%$, with the 16-h/lower-OLR configuration generally showing higher organic matter removal. In both strictly anaerobic and microaerated regimes, doubling the residence time from 8 h (R1 and R2) to 16 h (*R*3 and R4) resulted in the highest and most consistent removal rates (Fig. 3). This trend indicates that the 8-h HRT imposes kinetic limitations on the consortia, likely exacerbated by LAS inhibition, which hinders the full stabilization of the sucrose-based influent regardless of the aeration status.

Microaeration served as an effective oxidative boost, particularly at the shorter HRT. R2 (8 h, microaerated) achieved $77.6 \pm 6.8\%$, significantly outperforming its anaerobic counterpart (R1) ($p = 0.042$). This improvement can be attributed to the stimulation of facultative

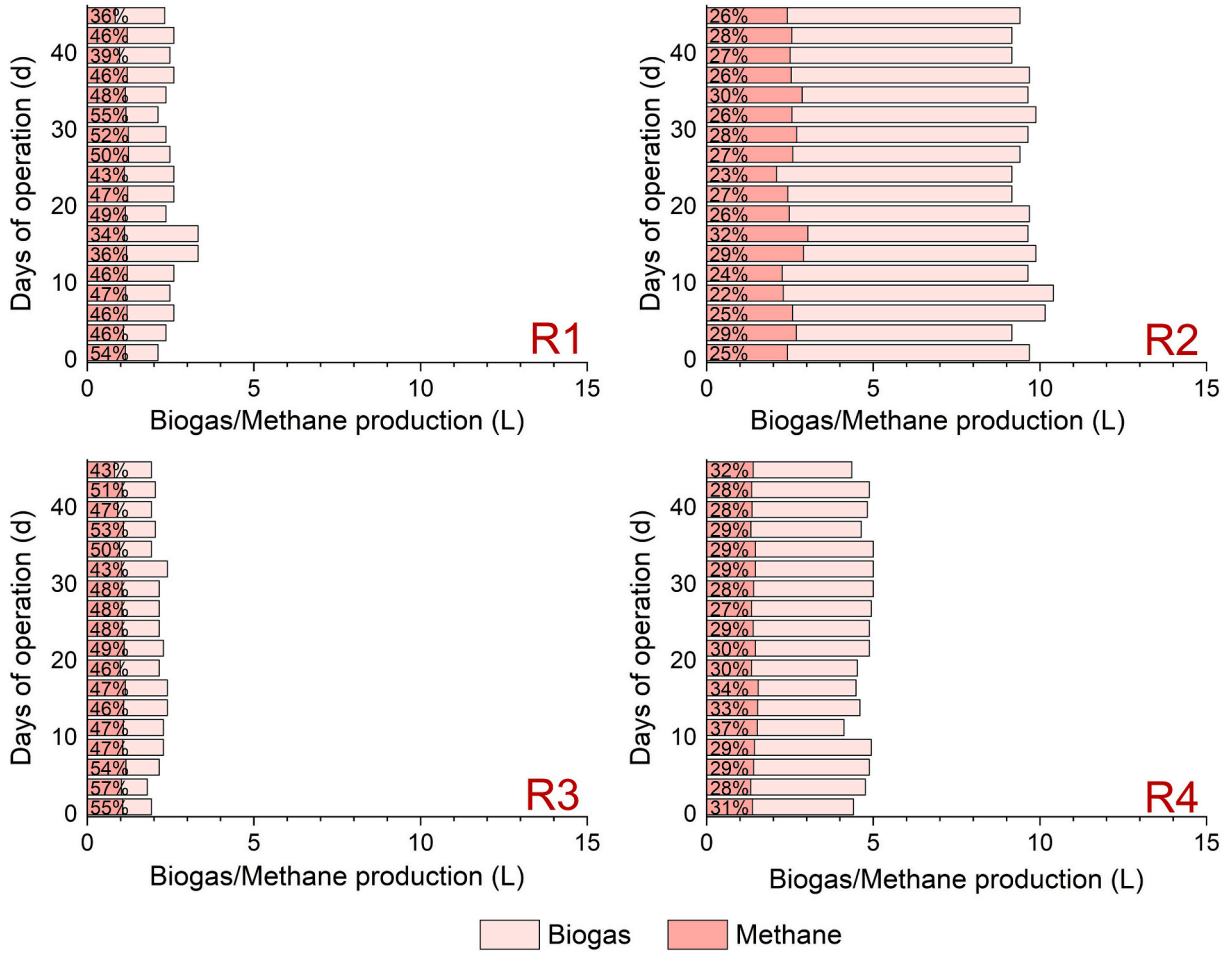


Fig. 2. Biogas and methane production from reactors throughout the experiment.

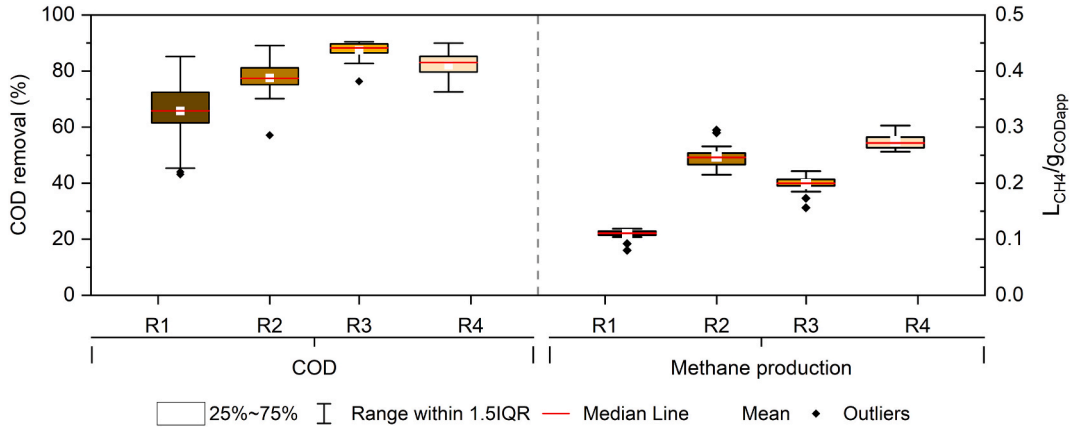


Fig. 3. Boxplot of COD removal and relative methane production.

microorganisms with faster kinetic rates for oxidizing organic intermediates (do Nascimento et al., 2021). At 16 h, microaeration did not significantly affect the degradation of bulk organic matter, whereas R4 (82.1%) and *R3* (87.4%) showed similar removal efficiencies ($p = 0.81$). These results suggest that while the 16-h HRT provided greater contact time for anaerobic stabilization than the 8-h HRT, microaeration was particularly important for maintaining system functionality and intensifying the process at higher volumetric loading rates.

Normalization of methane production by the applied COD load provided a different perspective than the volumetric production results, with the highest specific methane yield observed in the 16-h microaerated reactor (R4). While the strictly anaerobic reactors (R1 and *R3*) and the 8-h microaerated reactor (R2) showed average yields of 0.11 ± 0.01, 0.20 ± 0.02 and 0.24 ± 0.01 L_{CH4}/g_{CODapp}, respectively, R4 (16 h, microaerated) achieved a remarkable 0.28 ± 0.01 L_{CH4}/g_{CODapp}.

This performance significantly exceeds the yields reported by Duarte et al. (2010), who observed specific methane yields around 0.2 L_{CH4}/g_{CODapp} in UASB reactors treating LAS under strictly anaerobic conditions, where surfactant toxicity limited methanogenic efficiency. The higher methane yield compared to the anaerobic control (*R3*)

demonstrates that controlled oxygenation acts as a process intensifier. This trend is supported by Delforno et al. (2012), who reported that introducing oxygen-mediated stages into the treatment of anionic surfactants led to more robust aromatic ring degradation, thereby increasing the availability of acetate for methanogens and boosting specific methane production relative to purely anaerobic benchmarks. By overcoming the typical kinetic barriers of surfactant-rich wastewater, R4 maximized the conversion of the applied organic load into energy.

Although R2 exhibited almost twice the absolute methane production of R4 (2.6 vs. 1.4 $L_{CH4}{\cdot}d^{-1}$), normalization by the applied COD load reversed this relationship, with R4 showing a higher specific methane yield than R2. This distinction indicates that the greater volumetric methane production at the 8-h HRT was largely associated with its higher daily organic load. In contrast, the 16-h microaerated condition converted a greater fraction of the applied COD into methane.

The higher specific methane yield observed under microaeration suggests that limited oxygen supplementation did not compromise methanogenesis and may have favored the conversion of the applied organic load into methane. A plausible explanation is that microaeration facilitated initial transformations of recalcitrant organic compounds, potentially increasing the availability of intermediates for subsequent anaerobic metabolism (Chen et al., 2020). However, because intermediate products and metabolic fluxes were not directly characterized, the mechanisms underlying the enhanced methane yield cannot be conclusively established from the present data.

3.2. Removal of linear alkylbenzene sulfonate

LAS removal varied with operational conditions, with efficiencies ranging from 34.8% (R1) to 64.0% (R4) (Fig. 4). During the early operational phase, all four reactors achieved average LAS removal efficiencies exceeding 70%, with no statistically significant differences between the systems ($p = 0.65$). This performance is linked to the high functional diversity of the incipient anaerobic microbiota, which had not yet been subjected to the selective pressure of sulfonated compounds. At this stage, the likely minimal accumulation of toxic intermediates, combined with the absence of chronic stress, favored the activity of consortia capable of at least partially degrading xenobiotics. Furthermore, because LAS concentrations remained below inhibitory thresholds, microbial activity was not significantly suppressed, allowing broad, nonspecific metabolic pathways to participate in the initial co-metabolic transformation.

The chemical nature of the pollutant dictates the shift from initial transformation to long-term stability. Specifically, the highly stable sulfonated aromatic ring and the hydrophobic alkyl chain of LAS impose severe thermodynamic constraints under strictly anaerobic conditions (R1 and *R*3). In the absence of molecular oxygen, benzene ring cleavage is restricted to the multi-step benzoyl-CoA pathway. This slow reductive route is frequently hindered by the molecule's inherent toxicity (Fu et al., 2023), thus limiting the overall rate of organic matter stabilization.

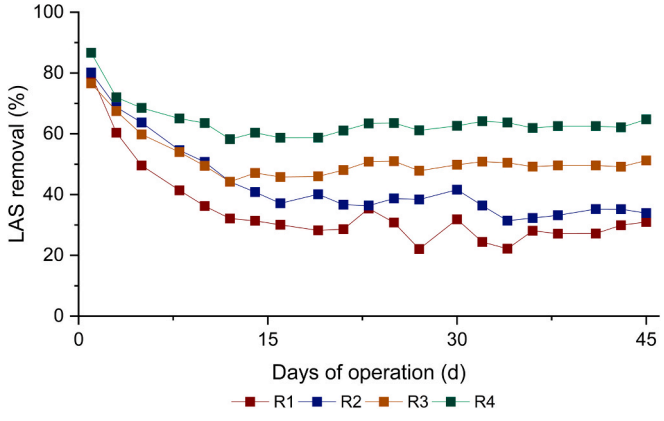


Fig. 4. LAS removal during the experiment.

In strictly anaerobic systems, doubling the HRT from 8 h (R1) to 16 h (*R*3) increased LAS removal from 34.8% to 51.8%. This performance gain is consistent with the hypothesis that the structural recalcitrance of the sulfonated aromatic group may require a longer kinetic window to accommodate the relatively slow metabolic adaptation of anaerobic consortia (Corada-Fernández et al., 2018). The limitation observed at 8 h aligns with the findings of Carneiro et al. (2024), who reported that short residence times (e.g., 6 h) in anaerobic stages are often insufficient for the effective biotransformation of complex aromatic rings, resulting in negligible removal of recalcitrant micropollutants. In these systems, cleavage of the benzene ring under strictly reductive conditions is the rate-limiting step. The longer contact time in the 16-h configuration may have helped alleviate mass-transfer and reaction-rate constraints that commonly limit degradation in high-rate anaerobic systems.

The introduction of microaeration was associated with a further increase in LAS removal, with the highest efficiency observed in R4 (64.0 ± 6.2%). While oxygen availability might have activated monooxygenases and dioxygenases in both microaerated reactors, the longer contact time in R4 may have favored the progression of oxygen-dependent transformation steps proposed for LAS degradation. According to previously described LAS degradation pathways, oxygen-dependent reactions may decrease the molecule's hydrophobicity through oxidative transformations that generate more hydrophilic intermediates, including sulfophenylcarboxylates (SPCs) (Liu et al., 2024; Zhou et al., 2022). Although these intermediates were not directly quantified in the present study, the higher LAS removal observed in R4 is consistent with the possibility that similar oxidative transformations occurred under microaerated conditions.

In contrast, the 8-h HRT reactors (R1 and R2) exhibited significant instability and lower removal efficiencies. In R1, the short HRT likely led to excessive LAS adsorption onto the sludge granules. This accumulation of surfactants increases cell membrane permeability and inhibits slow-growing methanogens (Braga et al., 2015). While microaeration in R2 improved LAS removal (43.8%) compared with R1, its lower performance relative to R4 suggests that the shorter contact time may have constrained the progression of LAS biotransformation. Conversely, the higher removal achieved in R4 is consistent with the possibility that the longer contact time favored the progression of successive transformation steps initiated under microaerated conditions.

Importantly, the effect of microaeration can be more directly assessed by comparing within each HRT/OLR-matched pair. At 8 h, LAS removal increased from 34.8% in R1 to 43.8% in R2, corresponding to an increase of 9.0 percentage points (25.9% relative increase). Similarly, at 16 h, LAS removal increased from 51.8% in *R*3 to 64.0% in R4, representing a 12.2 percentage-point increase (23.6% relative increase). The comparable relative improvements observed under the two operational regimes indicate a consistent positive response to microaeration. This trend was also maintained when performance was expressed as volumetric LAS removal rate, which increased from 15.7 to 19.7 mg $LAS{\cdot}L_{reactor}^{-1}{\cdot}d^{-1}$ at 8 h and from 11.7 to 14.4 mg $LAS{\cdot}L_{reactor}^{-1}{\cdot}d^{-1}$ at 16 h. Thus, although the individual contributions of HRT and OLR cannot be separated across the two hydraulic conditions, the matched comparisons provide direct evidence of improved LAS removal associated with microaeration at both HRT/OLR regimes.

Overall, the results indicate that microaeration consistently improved LAS removal within both HRT/OLR-matched pairs, while the 16-h operational configuration provided the highest removal efficiencies among the conditions evaluated. The highest LAS removal was achieved with microaeration at the 16-h HRT (R4). This enhanced performance may be explained by the combination of an initial oxygen-mediated transformation of LAS under microaerated conditions and the longer contact time provided by the 16-h HRT, which may favor the progression of subsequent biotransformation steps. Thus,

simultaneously addressing oxidative and kinetic limitations may enhance the degradation of recalcitrant surfactants under anaerobic-microaerophilic conditions.

3.3. Microbial dynamics during the removal of LAS

3.3.1. Microbial community richness and diversity

The analysis of richness and diversity indices reveals a clear influence of microaeration on microbial community complexity and structural stability within the UASB systems (Table S1).

According to Chao1 and ACE indices, reactors operated under microaerated conditions exhibited higher microbial richness than their conventional counterparts, indicating that low oxygen availability promoted the establishment of a broader range of taxa. Similarly, Shannon and Simpson indices indicated that microaerated systems supported higher diversity, reflecting a more even distribution of taxa and reduced dominance. This pattern suggests that microaeration may have increased redox heterogeneity, enabling the coexistence of facultative, strictly anaerobic, and oxygen-tolerant microorganisms, thereby increasing functional redundancy and metabolic versatility.

The inoculum exhibited intermediate richness and diversity, lower than those observed in the microaerated reactors but higher than in the conventional UASB systems. In contrast, its diversity indices were comparable to the microaerated condition, indicating a relatively balanced community. Upon reactor operation, conventional UASB conditions, particularly at short HRT, exerted strong selective pressure, reducing richness and favoring fast-growing fermentative taxa. In contrast, microaeration mitigated this effect by expanding redox niches and sustaining a broader range of functionally relevant microorganisms, including syntrophic bacteria and methanogens.

This structuring effect is further supported by the Venn diagram analysis, which demonstrates pronounced differentiation among communities (Fig. S2). The inoculum contained a large number of unique ASVs, reflecting its heterogeneous origin. At the same time, all reactors developed distinct assemblages with comparable numbers of exclusive ASVs, indicating rapid community reorganization driven by operational conditions. The progressive increase in unique ASVs across the reactor configurations highlights the influence of the different operational conditions in expanding ecological niche space and enabling the establishment of specialized populations associated with LAS transformation and methanogenic activity.

The limited number of ASVs shared among all systems underscores the strong selective pressure imposed by reactor operation. It indicates that LAS degradation depended on reactor-specific microbial assemblies rather than on a conserved core microbiome. In microaerated reactors, a higher proportion of exclusive ASVs reflects greater functional diversification and community resilience, consistent with their superior performance in LAS removal. Overall, these results demonstrate that

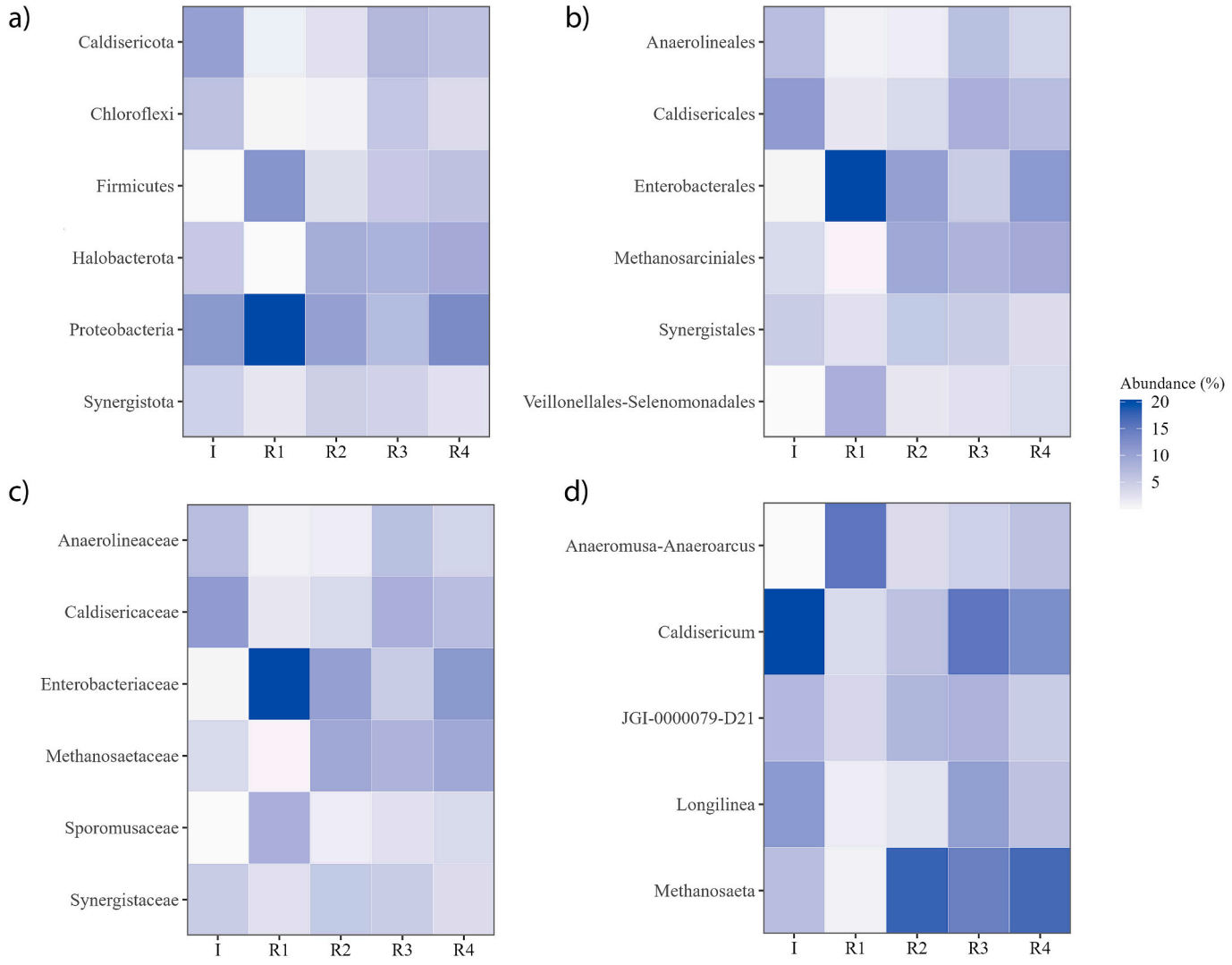


Fig. 5. Structure of the microbial community in terms of phylum, order, family, and genus.

microaeration, reinforced by longer HRT, promoted a more complex and functionally versatile microbial community, thereby enhancing the system capacity for efficient LAS transformation.

3.3.2. Microbial community structure

The microbial community structure differed among the hydraulic/loading and microaeration conditions evaluated (Fig. 5).

At the phylum level, *Proteobacteria* dominated across all systems, which aligns with their central role in the anaerobic transformation of complex organic compounds, including surfactants, via fermentative and facultative pathways (de Barros et al., 2018; Zhu et al., 2020). Relative to the inoculum, a reduction in *Caldisericota* and *Chloroflexi* was observed, particularly under the 8-h HRT, indicating that short contact times imposed strong selective pressure against slow-growing syntrophic and filamentous taxa involved in biomass structuring and long-chain organic matter conversion.

A comparison between the conventional UASB reactors operated at 8 h HRT (R1) and 16 h HRT (*R*3) highlights the role of residence time in shaping metabolic specialization and syntrophic stability. In R1, enrichment of *Firmicutes* alongside *Proteobacteria* may have favored fast fermentative metabolism, with dominance of *Enterobacterales*, *Veillonellales*, and *Selenomonadales*, and families associated with acidogenesis and the rapid conversion of readily biodegradable substrates into short-chain fatty acids and alcohols (do Nascimento et al., 2021; Zígolo et al., 2020).

The stimulation of *Anaeromusa–Anaeroarcus*, absent in the inoculum, further reflects selection for rapid fermentative pathways. Although such organisms may contribute to partial LAS transformation via co-metabolism and desulfonation, their metabolic orientation favors rapid carbon turnover rather than complete mineralization or stable syntrophic interactions with methanogens (Costa et al., 2024; Terreros-Mecalco et al., 2024). This configuration may help explain the lower LAS removal and moderate methane production in R1, along with inhibition of *Methanosaetaceae*, leading to sustained methane formation via less efficient fermentative and incomplete oxidation pathways.

In contrast, *R*3 exhibited a community structure indicative of enhanced process stability and syntrophic integration. The persistence of *Caldisericota* and enrichment of *Anaerolineales* reflect improved adaptation to slow hydrolysis, biomass structuring, and interspecies metabolite exchange. At the genus level, the co-occurrence of *Caldisericum*, *Methanosaeta*, and *Longilinea* indicates a tighter integration among fermentative, acetogenic, and acetoclastic methanogenic pathways. The greater abundance of *Methanosaetaceae* at the longer HRT suggests that this operational condition may have favored the establishment of these populations, which is particularly relevant for LAS degradation, where acetate is produced gradually through complex transformation pathways. Despite lower methane production than in R1, the higher LAS removal in *R*3 indicates greater progression of LAS transformation under the longer contact time.

Microaeration further intensified community differentiation, particularly under extended HRT. At 8-h HRT, R2 exhibited a mixed assemblage dominated by *Proteobacteria* and *Halobacterota*, with coexistence of facultative heterotrophs and metabolically flexible methanogens, including *Methanosarciniales*. This structure suggests that microaeration may have increased redox heterogeneity within the sludge bed, potentially supporting both oxidative and anaerobic metabolic functions while sustaining methane production despite the short residence time.

In R4, extending the HRT to 16 h under microaeration resulted in a markedly more favorable configuration. Stable enrichment of *Methanosarciniales* and *Methanosaetaceae*, together with *Anaerolineales*, suggests a microbial configuration potentially capable of supporting complementary facultative and anaerobic functions. The high abundances of *Methanosaeta* and *Caldisericum* are consistent with greater potential for integration among fermentative, acetogenic, and acetoclastic methanogenic processes, which may have contributed to the higher LAS removal observed in R4.

The persistence of strictly anaerobic methanogens in the microaerated reactors is particularly relevant considering that synthetic air was introduced directly at the bottom of the UASB reactors. Although this configuration exposed the sludge blanket to oxygen-containing gas, the maintenance of methanogenic populations and methane production indicates that oxygen supplementation was insufficient to suppress methanogenesis. A plausible explanation is the rapid consumption of the limited oxygen supply by facultative and oxygen-utilizing microorganisms, and the reduction of substrates in the lower sludge bed, thereby restricting oxygen penetration into the biomass. In addition, mass-transfer limitations within sludge aggregates may have contributed to the preservation of oxygen-depleted microenvironments in which strictly anaerobic methanogens could remain active. Thus, the coexistence of facultative populations and methanogens observed under microaeration is consistent with spatially heterogeneous oxygen availability within the sludge blanket. However, because spatial DO and ORP profiles were not measured, the establishment and distribution of these microenvironments cannot be directly confirmed from the present data.

Overall, comparisons within each HRT indicate that microaeration was associated with distinct microbial communities and greater LAS removal. Among the configurations evaluated, R4 achieved the highest LAS removal and exhibited a microbial profile with greater representation of taxa potentially involved in complementary fermentative, oxidative, and methanogenic functions.

3.3.3. Metabolic pathway analysis and functional implications

The distribution of metabolic pathways across the reactors reflected the selective pressures imposed by hydraulic retention time and microaeration and was fully consistent with the previously described microbial community structures (Fig. 6 and S3-S5).

In the conventional UASB operated at 8-h HRT, the dominance of sucrose metabolism, the TCA pathway, ω-oxidation, and sulfur metabolism indicates a system primarily oriented toward rapid fermentative activity and incomplete oxidation processes.

This functional profile is consistent with the enrichment of *Proteobacteria* and *Firmicutes*-related taxa, particularly *Enterobacteriaceae*, *Sporomusaceae*, and *Anaeromusa–Anaeroarcus*, which favor rapid substrate turnover but provide limited metabolic continuity for the complete degradation of complex molecules such as LAS (Carosia et al., 2014). Functional prediction suggested the presence of ω-oxidation pathways potentially associated with *Anaeromusa-Anaeroarcus* and *Rhodococcus*, indicating a possible capacity for initial alkyl chain activation.

In the microaerated UASB operated at 8-h HRT, the metabolic profile shifted toward pathways associated with improved energy conservation, with dominance of the Acetyl-CoA pathway and both hydrogenotrophic and acetoclastic methanogenesis. This configuration reflects a more organized metabolic sequence in which fermentative and acetogenic reactions were efficiently aligned with methane formation. *Methanosaeta* emerged as a central genus across all methanogenic routes, alongside *Methanobacterium* and *Methanospirillum*, which together facilitated efficient hydrogen consumption and acetate utilization.

Functional inference indicated that microaeration was associated with an increased genetic potential for ω-oxidation pathways involving *Longilinea*, *Desulfovibrio*, and *Anaeromusa-Anaeroarcus*, suggesting a possible enhancement in alkyl chain activation. Predicted β-oxidation pathways associated with *Pseudomonas* and *Acinetobacter* further indicate potential for processing of fatty acid intermediates. In addition, the inferred participation of *Thauera* and *Methanosaeta* in aromatic compound metabolism suggests a capacity to destabilize benzene rings. However, as these results are based on gene prediction rather than transcriptomic, proteomic, or metabolomic evidence, they represent potential metabolic capabilities that may help explain the higher LAS removal observed in R2 compared with the conventional reactor at the same HRT.

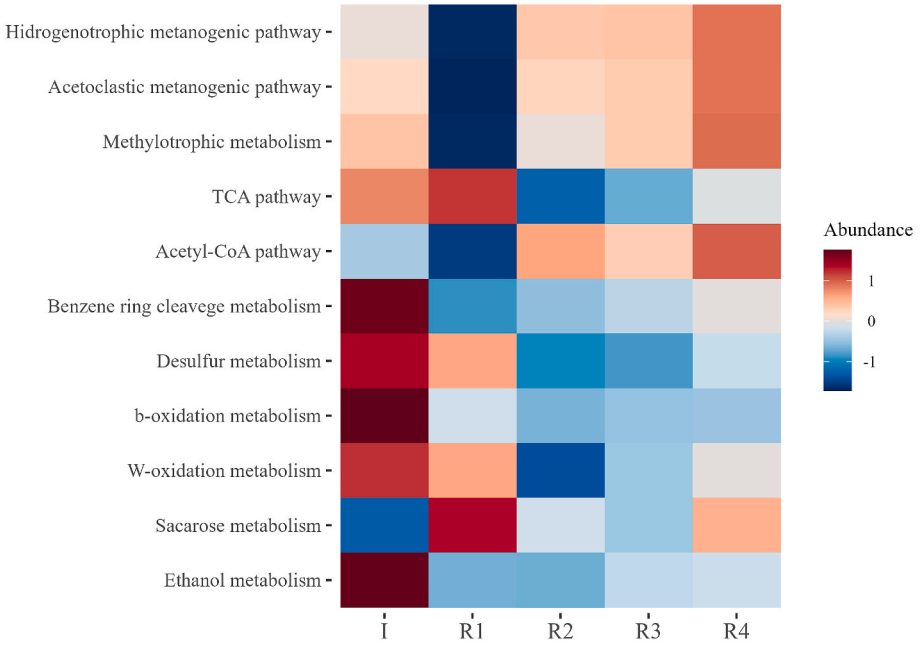


Fig. 6. Z-score of the relative abundance of metabolic pathways involved in the biological degradation process of LAS.

At 16-h HRT in the conventional UASB, functional prediction indicated enrichment of methanogenic pathways, including hydrogenotrophic, acetoclastic, and methylotrophic routes. The extended retention time was associated with a genetic profile consistent with a metabolically coordinated anaerobic network in which *Methanosaeta*, *Methanobacterium*, and *Methanolinea* potentially contributed to the turnover of hydrogen and acetate. Predicted ω-oxidation and sulfur metabolism pathways linked to *Longilinea* and *Desulfovibrio* suggest a genetic potential for alkyl chain activation and desulfonation under more stable conditions. Similarly, inferred β-oxidation pathways involving *Longilinea*, *Acinetobacter*, and *Brooklawnia* indicate a potential capacity for gradual conversion of intermediates, while predicted aromatic compound metabolism associated with *Thauera* and *Methanosaeta* suggests potential involvement in ring destabilization. These inferred metabolic capacities may explain the higher LAS removal in *R*3 compared to R1, although methane production likely remained constrained by kinetic limitations typical of longer HRT and acetoclastic dominance.

The microaerated UASB operated at 16-h HRT exhibited the most functionally diverse predicted metabolic profile. Functional inference indicated enrichment of the Acetyl-CoA pathway, along with genes associated with hydrogenotrophic, acetoclastic, and methylotrophic methanogenesis, suggesting a broad metabolic potential (Lara-Martín et al., 2010; Motteran et al., 2018). *Methanosaeta*, supported by *Methanobacterium* and *Methanolinea*, was associated with predicted pathways related to acetate and hydrogen utilization. Predicted ω-oxidation pathways linked to *Smithella*, *Tolumonas*, and *Longilinea* suggest a genetic capacity for alkyl chain activation. In contrast, inferred β-oxidation pathways associated with *Longilinea* and *Acinetobacter* indicate potential for intermediate processing.

In parallel, functional prediction suggested genetic potential for aromatic compound transformation (*Thauera* and *Methanosaeta*) and sulfur metabolism (*Desulfovibrio*). Together, these predicted capacities may help explain the superior LAS removal and stable methane production observed in R4.

Overall, the functional pathway prediction suggests that LAS removal in UASB systems may be associated with the degree of potential integration among fermentation, oxidation, desulfonation, and methanogenesis pathways. Short HRT was associated with a predicted metabolic profile consistent with rapid but potentially incomplete transformation routes. In contrast, longer HRT was associated with a functional signature indicative of more stable methanogenic pathways and greater potential for intermediate conversion. Microaeration was linked to an increased predicted genetic potential for early oxidative steps, particularly in the 16-h/lower-OLR configuration.

Furthermore, the observed microbial shifts provide indirect support for the coexistence of oxidative and anaerobic metabolic functions under microaeration. Nevertheless, microbial composition alone cannot establish the sequence of LAS transformation reactions. Direct characterization of LAS transformation products would be required to confirm whether the proposed oxidative and anaerobic pathways occurred under the conditions evaluated.

3.4. Gas-phase oxygen depletion in microaerated reactors using nitrogen as an internal tracer

Off-gas monitoring provided additional evidence of the fate of the oxygen supplied to the microaerated reactors. Residual O_2 remained low throughout operation, ranging from 0.20 to 1.22% in R2 and from 1.12 to 2.16% in R4 (Fig. S6). More importantly, the O_2/N_2 ratio remained consistently below 0.25 in the synthetic air supplied to both reactors. This distinction is relevant because CH_4 and CO_2 production dilutes O_2 and N_2 simultaneously and, in the absence of preferential O_2 depletion, would not substantially alter their ratio. The consistently lower O_2/N_2 values therefore indicate selective depletion of O_2 during reactor passage rather than merely dilution of the inlet gas by biologically produced gases.

This behavior resulted in high apparent gas-phase O_2 depletion throughout the experimental period, averaging 82.9 ± 5.7% in R2 and 83.1 ± 3.3% in R4 (Fig. S7). The similar depletion observed at both HRTs indicates that the supplied oxygen was extensively depleted under both microaerated configurations despite their different hydraulic and organic loading conditions. Importantly, this high O_2 depletion occurred without suppression of methanogenesis. R2 and R4 maintained methane production of approximately 2.6 and 1.4 L CH_4 d^{-1}, respectively, while their specific methane yields reached 0.24 ± 0.01 and 0.28 ± 0.01 L CH_4 $gCODapp^{-1}$. The simultaneous occurrence of extensive O_2 depletion and sustained methane formation supports the establishment of limited oxygen availability rather than unrestricted aerobic conditions within the microaerated systems.

The characterization and control of very low oxygen availability in anaerobic systems remains analytically challenging. ORP-based control represents an important approach for this purpose because changes in redox potential can provide an indirect response to oxygen addition even when dissolved O_2 concentrations are below the reliable detection range of conventional probes. Nguyen et al. (2019), for example, developed an intermittent microaeration strategy in which O_2 injection was controlled by increasing the ORP by 25 mV relative to the corresponding anaerobic baseline. This approach was subsequently applied to characterize microbial adaptation under microaerobic conditions (Wu et al., 2021) and, more recently, to control in-situ microaeration during anaerobic co-digestion (Chuenchart et al., 2025). Importantly, ORP reflects the overall redox state of the medium and is influenced not only by O_2 availability but also by electron acceptors and reducing equivalents as well as operational conditions such as pH, conductivity, and temperature (Wu et al., 2021). Therefore, ORP monitoring would have provided valuable complementary information on the redox conditions established within the sludge bed, and its absence is recognized as a limitation of the present study.

Gas-phase monitoring has also been used as an operational indicator of microaeration in anaerobic digesters, with airflow adjusted according to residual O_2 concentrations in the produced gas (Khadir et al., 2025). Here, rather than inferring a specific ORP or dissolved O_2 condition, the controlled low air supply was complemented by direct chromatographic monitoring of O_2 and N_2 in the off-gas. The consistently low residual O_2 concentrations and marked decrease in the O_2/N_2 ratio relative to the synthetic-air reference provide independent evidence that most of the supplied O_2 was depleted during reactor passage.

The gas-phase evidence is also consistent with the microbial structure observed in R2 and R4, where facultative and potentially oxygen-utilizing populations coexisted with methanogenic archaea, including *Methanosaeta* and members of *Methanosarciniales*. Rapid O_2 utilization in oxygen-exposed regions of the sludge bed, combined with mass-transfer limitations within the granular biomass, may have preserved oxygen-depleted niches capable of sustaining methanogenesis. Similar coexistence between facultative bacteria and methanogenic archaea has been reported under ORP-controlled microaeration, where limited O_2 exposure promoted functional adaptation without suppressing methane-producing populations (Nguyen et al., 2019; Wu et al., 2021). Such spatial and functional coexistence provides a plausible ecological explanation for the increase in LAS removal observed under microaeration, from 34.8 to 43.8% at 8 h and from 51.8 to 64.0% at 16 h, while anaerobic carbon conversion was maintained. Taken together, the controlled oxygen input, extensive gas-phase O_2 depletion, sustained methanogenesis, and microbial community structure support the establishment of limited oxygen availability within the microaerated reactors. Nevertheless, because ORP and dissolved O_2 were not directly monitored within the sludge bed, the spatial distribution and absolute redox conditions established inside the reactors cannot be resolved from the present dataset.

4. Conclusions

Microbial community analysis showed that the highest LAS removal (64%) in R4 was associated with a distinct microbial community characterized by enrichment of facultative and microaerophilic genera, including *Thauera* and *Desulfovibrio*, along with acetoclastic methanogens such as *Methanosaeta*. Functional predictions further indicated greater representation of oxygenase-related genes under microaeration, suggesting the potential for oxidative transformations to complement downstream anaerobic metabolism. These predictions, however, represent genetic potential rather than direct evidence of pathway expression or enzymatic activity.

Overall, microaeration combined with the longer HRT evaluated (16 h) provided the best LAS removal among the tested configurations while maintaining methanogenesis, demonstrating the potential of controlled oxygen supplementation to intensify predominantly anaerobic treatment of LAS-containing wastewater. From an implementation perspective, the results indicate that microaeration may be integrated into anaerobic treatment as a complementary strategy to enhance surfactant biotransformation without replacing the anaerobic process and its associated energy-recovery potential. Nevertheless, because HRT and OLR varied concomitantly between the 8-h and 16-h conditions, the superior performance of R4 should be interpreted as the response of the overall operational configuration rather than an isolated effect of HRT.

Although the combined reactor performance and microbial data support enhanced LAS biotransformation under microaeration, the lack of direct characterization of transformation products precludes definitive reconstruction of the metabolic pathway. Future studies coupling targeted or untargeted mass spectrometry with microbial functional analyses are therefore needed to identify LAS intermediates and confirm the transformation pathways suggested by the present results.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material containing alpha-diversity indices, pH and acidity/alkalinity data, microbial community and predicted metabolic pathway analyses, and off-gas oxygen profiles and depletion data.

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2026.135794.

Data availability

Data will be made available on request.

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